

石嘴山市华岳新材料科技有限公司

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技术数据表 • TECHNICAL DATA SHEET

氰氨化钙（石灰氮） • CALCIUM CYANAMIDE (Nitrolime) • 文件编号 Doc. No. HY-TDS-CCN-001

1. 产品标识 Product Identification

产品名称 Product name	氰氨化钙（工业级） Calcium cyanamide (industrial grade)	CAS 号 CAS No.	156-62-7
别名 Synonyms	石灰氮、氰氨基化钙 Nitrolime, lime nitrogen, calcium carbimide	EC 号 EC No.	205-861-8
分子式 Molecular formula	CaCN ₂	分子量 Molecular weight	80.10 g/mol
产品标准 Product standard	HG/T 5922-2021 《工业氰 氨化钙》 HG/T 5922-2021 Calcium cyanamide for industrial use	海关编码 HS code	3102 10 00 / 2851 00 00 (以报关口岸核定为准)

2. 外观与物化性质 Appearance & Physicochemical Properties

外观 Appearance	灰黑色至深灰色粉末或颗粒 Grey-black to dark grey powder or granules
熔点 Melting point	约 1145–1217 °C（升华 / 分解） approx. 1145–1217 °C (sublimes / decomposes)
堆积密度 Bulk density	约 0.8–1.1 g/cm ³ （视粒度与压实状态） approx. 0.8–1.1 g/cm ³ , depending on particle size and compaction
pH 值 pH	约 4.8（1 % 水悬浮液，水解后呈强碱性） approx. 4.8 (1 wt% aqueous suspension); strongly alkaline after hydrolysis
水中行为 Behaviour in water	遇水缓慢水解，生成单氰胺与氢氧化钙，并可能释放乙炔等可燃气体 Hydrolyses slowly in water to cyanamide and calcium hydroxide; may release acetylene and other flammable gases
溶解性 Solubility	不溶于有机溶剂；与水反应 Insoluble in organic solvents; reacts with water

3. 技术指标 Technical Specification — HG/T 5922-2021

序号 No.	项目 Item	单位 Unit	优等品 Superior grade	一等品 First grade	合格品 Qualified grade
1	有效氮含量 Available nitrogen content	%	≥ 21.5	—	—
2	碳化钙（电石）含量 Calcium carbide content	%	≤ 0.2	—	—
3	筛余物（180 μm 试验筛）质量分数 Sieve residue (180 μm test sieve), mass fraction	%	≤ 10	—	—

注：优等品指标依据 HG/T 5922-2021 并经第三方检验确认；一等品与合格品栏请按贵司持有的标准文本填写后发布。

Note: Superior-grade limits are per HG/T 5922-2021 and confirmed by third-party testing. First-grade and Qualified-grade cells are to be

completed from the company's controlled copy of the standard before release.

4. 典型批次检验结果 Typical Batch Test Results

项目 Item	技术要求 Requirement	实测值 Result	单项判定 Verdict
有效氮含量 Available nitrogen	≥ 21.5 %	22.1 %	合格 Pass
碳化钙含量 Calcium carbide	≤ 0.2 %	0.2 %	合格 Pass
筛余物 (180 μm) Sieve residue (180 μm)	≤ 10 %	3 %	合格 Pass

数据来源：宁夏计量质量检验检测研究院检验报告 №24092633，样品编号 24092633，生产日期 2024-06-07，检验日期 2024-06-26 至 2024-07-01，样品等级：优等品。检验结论：该样品所检项目符合标准要求。
Source: Test Report No. 24092633 issued by Ningxia Institute of Metrology, Quality Inspection and Testing. Sample No. 24092633; production date 2024-06-07; testing 2024-06-26 to 2024-07-01; sample grade: Superior. Conclusion — all tested items conform to the standard.
典型值仅代表该批次，不构成规格保证；以随货检验报告为准。Typical values represent that batch only and do not constitute a specification guarantee; the batch certificate of analysis governs.

5. 主要用途 Principal Applications

化工中间体 Chemical intermediate	生产单氰胺、双氰胺、胍盐、肌酸、硝酸胍及含氮衍生物的基础原料 Feedstock for cyanamide, dicyandiamide, guanidine salts, creatine, guanidine nitrate and other nitrogen derivatives
冶金 Metallurgy	钢铁脱硫脱磷剂、渗氮处理剂 Desulphurising / dephosphorising agent for steel; nitriding agent
农业 Agriculture	缓释氮肥、土壤调理与消毒、除草抑菌（须按当地农化登记法规使用） Slow-release nitrogen fertiliser, soil conditioning and disinfection, herbicidal/fungistatic use — subject to local agrochemical registration
其他 Others	水泥外加剂、废水脱硫及重金属处理 Cement additive; wastewater desulphurisation and heavy-metal treatment

6. 包装、贮存与运输 Packaging, Storage & Transport

包装 Packaging	25 kg / 50 kg 内衬聚乙烯薄膜编织袋；1000 kg 内衬防潮吨袋；可按客户要求定制 25 kg / 50 kg PE-lined woven bags; 1000 kg moisture-barrier jumbo bags; custom packing on request
贮存 Storage	存放于阴凉、干燥、通风良好的库房，远离火种、热源、水及潮气；与酸类、氧化剂、水源分开存放；容器保持密封 Store in a cool, dry, well-ventilated warehouse away from ignition sources, heat, water and moisture; segregate from acids, oxidisers and water; keep containers tightly closed
保质期 Shelf life	原包装密封干燥条件下 12 个月 12 months in sealed original packaging under dry conditions
运输 Transport	UN 1403，第 4.3 类 遇水放出易燃气体的物质，包装类别 III；须防雨防潮，禁止与酸类、氧化剂混装混运 UN 1403, Class 4.3 (substances which in contact with water emit flammable gases), PG III; protect from rain and moisture; do not load with acids or oxidisers

安全提示 / Safety notice: 本品遇水放出易燃气体，吞咽有害，造成严重眼损伤及皮肤刺激，可能引起皮肤过敏和呼吸道刺激。使用前请阅读本品安全技术说明书（SDS）。摄入酒精后接触本品可能引起面部潮红等类双硫仑反应。
This product releases flammable gases in contact with water, is harmful if swallowed, causes serious eye damage and skin irritation, and may cause skin sensitisation and respiratory irritation. Read the Safety Data Sheet before use. Alcohol consumption in connection with exposure may cause an antabuse-like flushing reaction.

守当地法律法规。

The information herein is based on our present knowledge, is given in good faith for guidance only, and constitutes neither a warranty of quality nor a representation of fitness for any particular purpose. Users must verify suitability for their own application and comply with all applicable laws and regulations.